



Step 7

Upper Elementary /
Early Middle School



Today I am learning to use letters and symbols to represent unknowns and express rules in patterns.

1. What Does the Box Mean?

Find the missing number.

a) $3 + \square = 8$

Answer: _____

b) $10 - \square = 4$

Answer: _____

c) $\square + 7 = 12$

Answer: _____

d) $15 = \square + 6$

Answer: _____

2. Replace the Box with a Letter

The letter stands for
the missing number.

a) $n + 3 = 8$

$n =$ _____

b) $12 - p = 5$

$p =$ _____

c) $m + 10 = 15$

$m =$ _____

d) $20 = r + 4$

$r =$ _____

3. What Is the Rule?

Look at the pattern.

Input	1	2	3	4
Output	3	4	5	6

Circle the rule.

Add 1

Add 2

Add 3

Write the rule: Output = Input + _____

4. Use a Letter Rule

Rule: $n + 2$

Complete the table.

n	1	2	3	4	5
Answer					

5. Which Number Does the Letter Stand For?

a) $a + 5 = 11$

$a =$ _____

b) $b + 4 = 9$

$b =$ _____

c) $c - 2 = 7$

$c =$ _____

d) $d + 8 = 20$

$d =$ _____

6. Pattern Rules

Look at the table.

Input	1	2	3	4
Output	5	7	9	11

Write the rule using a letter.

Output = $n +$ _____

7. Create Your Own Rule

Choose a rule.

Rule: _____

Complete your table.

Input	1	2	3	4
Output				



8. Think and Explain



a) Why do mathematicians use letters?



b) What does a variable mean?



c) How can a rule help us predict the next number?



I can...

- ☐ use symbols and letters for unknowns.
- ☐ write simple rules using letters.
- ☐ find missing values using rules.
- ☐ explain how rules work in patterns.



My Learning Goal

I can use letters and symbols
to show unknowns and rules
in number patterns.

